Power BI Assignment 1 – Data Transformation & Data Modeling

Tasks

Data Import & Transformation

1. Data Import:

○ Import List of Orders.csv into Power BI.

○ Open the list of orders in the Power Query Editor using the Transform Data option.

○ Import Order Details.csv and Sales Target.csv into Power Query Editor.

Answer:

Data Import in Power BI

Step 1: Import List of Orders.csv

- Open **Power BI Desktop**.
- Click **Get Data → Text/CSV**.
- Select **List of Orders.csv**.
- Click **Transform Data**.

Step 2: Import Order Details.csv

- In Power Query Editor, click **New Source**.
- Select **Text/CSV**.
- Choose **Order Details.csv**.
- Click **OK**.

Step 3: Import Sales Target.csv

- Click **New Source** again.
- Select **Text/CSV**.
- Choose **Sales Target.csv**.
- Click **OK**.

Now, all **three CSV files** are imported into the **Power Query Editor**.\

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2. Row Limiting & Data Types:

○ Restrict the List of Orders table to the first 500 rows.

○ Set the Order Date column to the Date data type.

○ Change the Amount and Target columns to fixed decimal numbers.

Answer:

Keep the first 500 rows

- Select the **List of Orders** table.
- Go to **Home → Keep Rows → Keep Top Rows**.
- Enter **500**.
- Click **OK**.

Change Order Date to Date

- Select the **Order Date** column.
- Right-click the column heading.
- Select **Change Type → Date**.
- If you get a date error, use **Change Type → Using Locale** and select **English (India)**.

Change Amount to Fixed Decimal Number

- Select the **Amount** column.
- Right-click → **Change Type → Fixed Decimal Number**.

Change Target to Fixed Decimal Number

- Select the **Target** column.
- Right-click → **Change Type → Fixed Decimal Number**.

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3. Text Formatting:

○ Format the Customer Name column to proper case to ensure consistent capitalization.

Answer:

- Select the **Customer Name** column in Power Query Editor.
- Go to **Transform → Format → Capitalize Each Word**.

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4. Column Creation:

○ Merge the City and State columns to create a new column named Location in the format:

City, State

○ Create a custom column named Profit Margin calculated as: Profit / Amount (expressed as

a percentage)

Answer:

1. Create Location Column:

- Select **City** and **State** columns.
- Go to **Add Column → Merge Columns**.
- Choose **Comma (,)** as the separator.
- Name the new column **Location as State and City**.

2. Create Profit Margin Column:

- Go to **Add Column → Custom Column**.
- Name it **Profit Margin**.
- Enter the formula:

```
[Profit] / [Amount]
```

- Click **OK** and change the result to **Percentage**.

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5. Conditional Column:

○ Add a conditional column named Profit Status based on:

■ Profit < 0 → Loss

■ Profit = 0 → Break-Even

■ Profit > 0 → Profit

Answer:

Go to **Add Column → Conditional Column**.
Name the column **Profit Status**.
Set the conditions:

- If **Profit** is less than 0 → **Loss**
- If **Profit** equals 0 → **Break-Even**
- If **Profit** is greater than 0 → **Profit**

Click **OK**.

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6. Merging Data (Joins)

○ Merge the List of Orders and Order Details tables using Order ID.

○ Name the merged output table as Orders Data.

Answer:

- In **Power Query Editor**, select the **List of Orders** table.
- Go to **Home → Merge Queries**.
- Select **Order Details** as the second table.
- Select the **Order ID** column in both tables.
- Click **OK**.
- Expand the merged column to add the required details.
- Rename the merged table as **Orders Data**.

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7. Handling Missing Data & Duplicate Data

● Identify missing values across all tables.

● Define and apply appropriate strategies such as:

○ Replacing missing values

○ Removing records

○ Retaining nulls with justification

● Identify duplicate rows and define a suitable handling strategy based on business logic.

Answer:

- Select the **Order ID** column.
- Go to **Home → Remove Rows**.
- Click **Remove Duplicates**.

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8. Sorting and Filtering Data

● Sort records in the Orders Data table by Order Date in descending order.

● Apply filters to analyze orders from a specific state (e.g., Tamil Nadu).

Answers:

Sort by Order Date

1. Select the **Order Date** column in the **Orders Data** table in sales target.
2. Click the **sort arrow** in the column header.
3. Select **Descending**.

Filter by State

1. Select the **State** column.
2. Click the **filter dropdown**.
3. Select **Tamil Nadu**.
4. Click **OK**.

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9. Grouping and Aggregating Data

● Duplicate the Order Details table and perform the following:

○ Count of Order ID

○ Average Profit by Category

○ Total Amount by Sub-Category

● Duplicate the Sales Target table and aggregate the total Target by Month of order date.

Answer:

A. Duplicate Order Details Table

Step 1: Duplicate the Table

1. Right-click the Order Details table.
2. Click Duplicate.

Step 2: Count of Order ID

1. Select the duplicated table.
2. Go to Home → Group By.
3. Select Order ID in Group By.
4. Select Count Rows in Operation.
5. Click OK.

Step 3: Average Profit by Category

1. Right-click Order Details → Duplicate.
2. Go to Home → Group By.
3. Select Advanced.
4. Group by Category.
5. New column name: `Average Profit`.
6. Operation: Average.
7. Column: Profit.
8. Click OK.

Step 4: Total Amount by Sub-Category

1. Right-click Order Details → Duplicate.

2. Go to Home → Group By.
3. Select Advanced.
4. Group by Sub-Category.
5. New column name: `Total Amount`.
6. Operation: Sum.
7. Column: Amount.
8. Click OK.

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10. Data Modeling:

● Establish a relationship between:

○ List of Orders and Order Details using Order ID

● Establish a relationship between:

○ Order Details and Sales Target using Category

2. Use Manage Relationships to ensure relationships are active and correctly defined.

Answer:

- Open Model View in Power BI.
- Click Manage Relationships → New.
- Connect List of Orders and Order Details using Order ID.
- Set the relationship to One-to-Many and click Save.
- Click New again.
- Connect Order Details and Sales Target using Category.
- Make both relationships Active.
- Click Save and Close.
